

YEAR 10 SCIENCE  
SEMESTER 2 EXAMINATION, 1994  
PHYSICS 1 AND 2

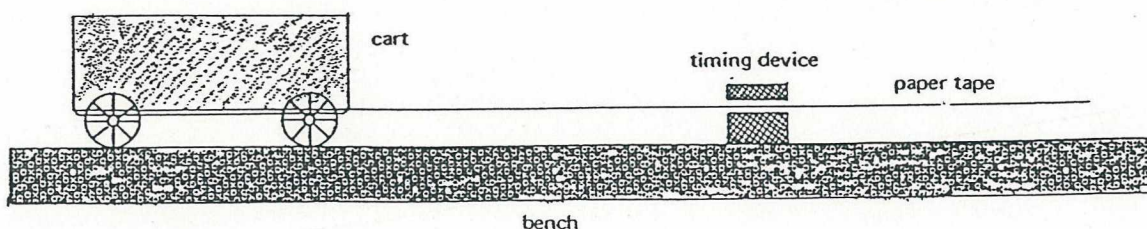
**MULTIPLE CHOICE**

**CONSTANTS:** Acceleration due to gravity  $g = 9.80 \text{ ms}^{-2}$

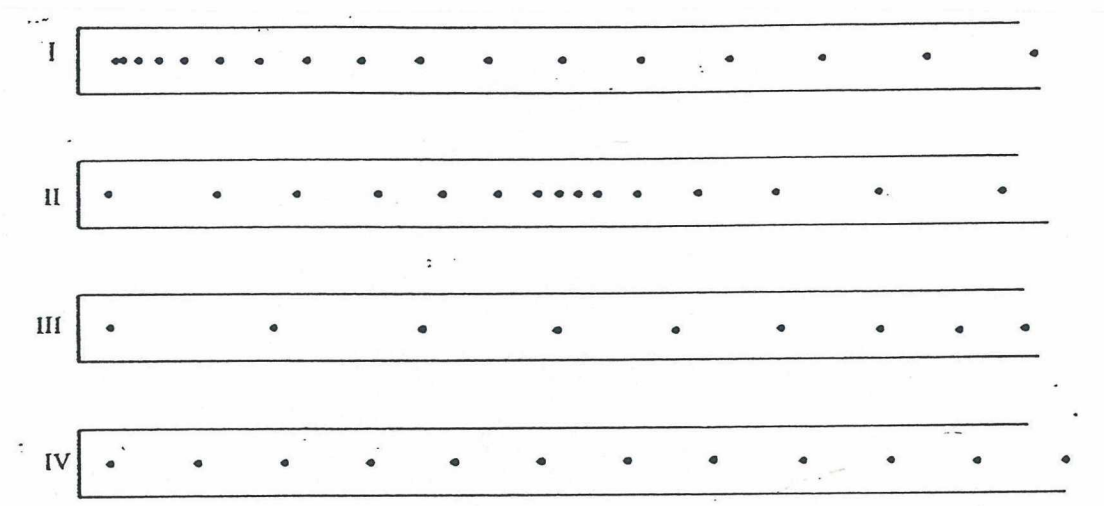
1. Which of the following is a scalar quantity?  
(a) velocity                      (b) acceleration                      (c) energy                      (d) force
2. Which one of the following masses is expressed to three significant figures?  
(a) 0.0610                      (b) 3.275                      (c) 0.00069                      (d) 6.370
3. An aeroplane is travelling from airport 1 to airport 2, which is due west of airport 1. For most of its flight the plane encounters a steady wind from the south. For the plane to travel along a straight line path from airport 1 to airport 2, the pilot should head the nose of the plane along a compass bearing which is:  
(a) due west.  
(b) due south.  
(c) between west and north.  
(d) between west and south.
4. A car has an initial velocity of  $15.0 \text{ ms}^{-1}$  east and uniformly accelerates to  $25.0 \text{ ms}^{-1}$  in 4.00 s. How far has the car moved during this change of velocity?  
(a)  $1.40 \times 10^2 \text{ m}$                       (b) 50.0 m                      (c) 80.0 m                      (d) 65.0 m

THE NEXT 3 QUESTIONS REFER TO THE FOLLOWING INFORMATION.

A cart moves along a level bench, and is attached to the cart is a long strip of paper which is drawn through a ticker-timer that records 50 dots per second.



The following diagrams represent the marks made on four separate tapes by the timing device as the cart moves.



5. Which one of the diagrams best indicates that the cart was slowing down throughout the recorded time interval?  
 (a) I (b) II (c) III (d) IV
6. Which one of the diagrams best indicates that the cart started from rest and accelerated?  
 (a) I (b) II (c) III (d) IV
7. Which one of the diagrams best indicates that the cart was travelling at constant speed?  
 (a) I (b) II (c) III (d) IV

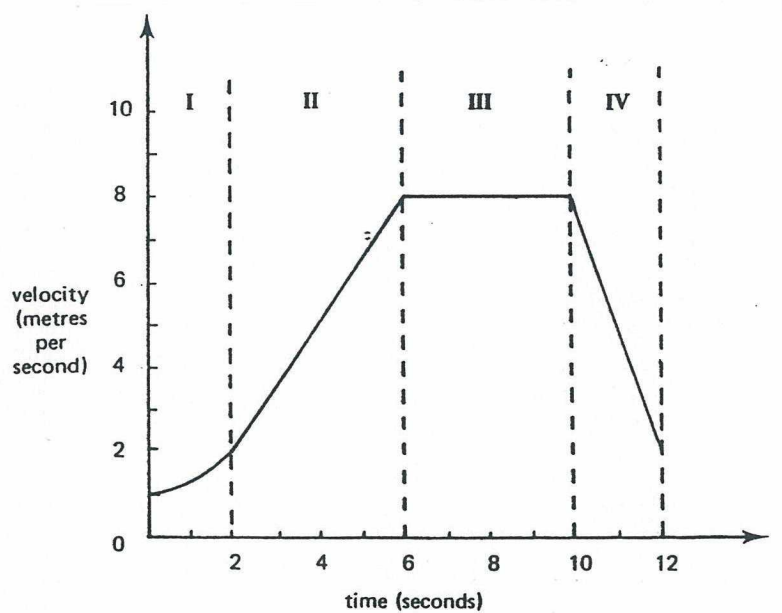
THE NEXT 3 QUESTIONS REFER TO THE FOLLOWING INFORMATION.

A body falls from rest near the Earth's surface. Ignore the effects of any air resistance.

8. The velocity of the body after 1.00 s of fall is:  
 (a)  $4.90 \text{ ms}^{-1}$  (b)  $9.80 \text{ ms}^{-1}$  (c)  $19.6 \text{ ms}^{-1}$   
 (d) unable to be calculated from the information.
9. The distance the body falls in the first 2.00 s is:  
 (a) 4.90 m (b) 9.80 m (c) 19.6 m  
 (d) unable to be calculated from the information.
10. If at a certain instant the velocity of the body is  $29.4 \text{ ms}^{-1}$ , its velocity 1.00 s later is:  
 (a)  $29.4 \text{ ms}^{-1}$  (b)  $34.3 \text{ ms}^{-1}$  (c)  $39.2 \text{ ms}^{-1}$  (d)  $44.1 \text{ ms}^{-1}$
11. A car has an initial velocity of  $20.0 \text{ ms}^{-1}$  and slows uniformly to  $5.00 \text{ ms}^{-1}$  in 4.00 s. In this time it travels:  
 (a) 25.0 m (b) 50.0 m (c)  $1.00 \times 10^2 \text{ m}$  (d)  $2.00 \times 10^2 \text{ m}$

THE NEXT 3 QUESTIONS REFER TO THE FOLLOWING INFORMATION.

The velocity-time graph below illustrates the motion of a body moving in a straight line.



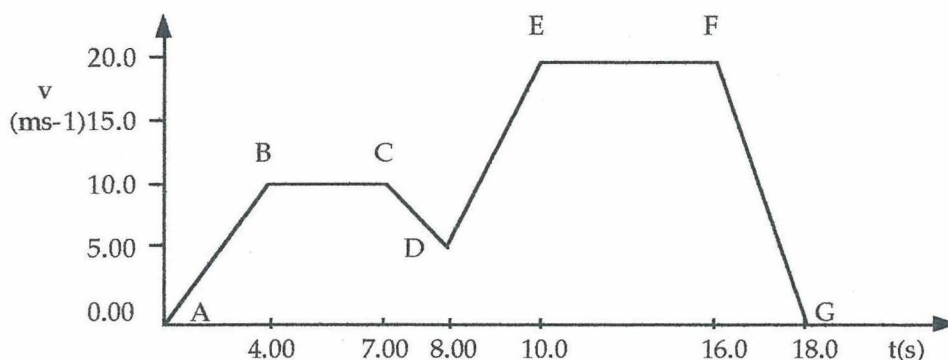
12. During which of the periods I, II, III and IV, was the magnitude of the average acceleration greatest?
- (a) I                      (b) II                      (c) III                      (d) IV
13. When was the object stationary?
- (a) At the start of period I.  
(b) During period III.  
(c) At the end of period IV.  
(d) At none of these times.
14. During which of the periods did the object travel the greatest distance?
- (a) I                      (b) II                      (c) III                      (d) IV
15. A SONAR impulse (a sound wave) is transmitted vertically downward from a stationary boat and the signal is reflected from the sea bed. It is received on the boat 10.0 s after the impulse is sent. Given that the speed of sound in the water is  $1.45 \times 10^3 \text{ ms}^{-1}$ , how deep is the water in this part of the ocean?
- (a) 725 m                      (b)  $1.45 \times 10^3 \text{ m}$                       (c)  $7.25 \times 10^3 \text{ m}$                       (d)  $1.45 \times 10^4 \text{ m}$

## SHORT-ANSWER QUESTIONS

Answer all questions. Credit will be given for incorrect answers provided your working is clearly and legibly set out.

1. A ticker timer that records 50 dots per second was used to produce the following tape during an experiment. Analyse the tape to determine the acceleration represented. (4)
- 

2. The movement of an object is represented in the following graph.



- (a) Describe the type of motion taking place in sections AB, CD, EF and FG. (2)  
(b) Calculate the greatest acceleration that takes place. (2)  
(c) How far does the object move? (3)
3. A train moving between Whitfords and Warwick stations moves as follows: it accelerates from rest at Whitfords to reach  $1.10 \times 10^2 \text{ kmh}^{-1}$  in 14.0 s, and maintains this velocity until it stops in a distance of  $1.80 \times 10^2 \text{ m}$  at Warwick. Determine the acceleration in both cases. (5)
4. In Western movies, crowds of people are sometimes seen firing rifles and hand-guns into the air as a sign of jubilation. Given that a rifle has a muzzle velocity of  $3.90 \times 10^2 \text{ ms}^{-1}$  and the bullet is fired straight up, calculate:  
(a) the maximum height reached by the bullet.  
(b) the time of flight.  
(c) the velocity with which it returns to the height from which it is fired. (5)
5. A plane flying at  $4.00 \times 10^2 \text{ kmh}^{-1}$  N  $20.0^\circ$  W alters course to fly W  $30.0^\circ$  S at  $4.50 \times 10^2 \text{ kmh}^{-1}$ . What is the change in velocity of the plane? (4)
6. A person runs  $1.50 \times 10^2 \text{ m}$  E in 25.0 s and then  $1.80 \times 10^2 \text{ m}$  S in 29.0 s. Calculate:  
(a) the displacement of the person.  
(b) the average speed for the motion.  
(c) the average velocity for the person. (6)

NAME: SEM 2 EXAM, 1994

SUBJECT: PHYSICS

MARK THE LETTER WHICH CORRESPONDS TO THE MOST APPROPRIATE ANSWER  
WITH A CROSS

|    |              |              |              |              |   |
|----|--------------|--------------|--------------|--------------|---|
| 1  | A            | B            | <del>C</del> | D            | E |
| 2  | <del>A</del> | B            | C            | D            | E |
| 3  | A            | B            | C            | <del>D</del> | E |
| 4  | <del>A</del> | B            | C            | D            | E |
| 5  | A            | B            | <del>C</del> | D            | E |
| 6  | <del>A</del> | B            | C            | D            | E |
| 7  | A            | B            | C            | <del>D</del> | E |
| 8  | <del>A</del> | B            | C            | D            | E |
| 9  | A            | B            | <del>C</del> | D            | E |
| 10 | A            | B            | <del>C</del> | D            | E |
| 11 | A            | <del>B</del> | C            | D            | E |
| 12 | A            | B            | C            | <del>D</del> | E |
| 13 | A            | B            | C            | <del>D</del> | E |
| 14 | A            | B            | <del>C</del> | D            | E |
| 15 | A            | B            | <del>C</del> | D            | E |
| 16 | A            | B            | C            | D            | E |
| 17 | A            | B            | C            | D            | E |
| 18 | A            | B            | C            | D            | E |
| 19 | A            | B            | C            | D            | E |
| 20 | A            | B            | C            | D            | E |

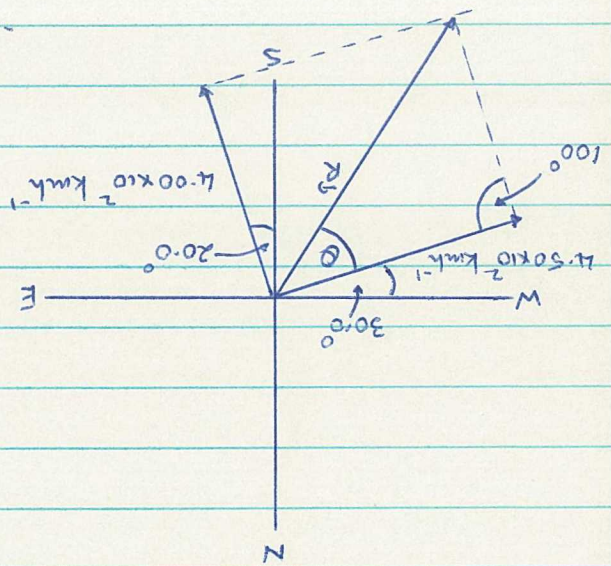
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| 21 | A | B | C | D | E |
| 22 | A | B | C | D | E |
| 23 | A | B | C | D | E |
| 24 | A | B | C | D | E |
| 25 | A | B | C | D | E |
| 26 | A | B | C | D | E |
| 27 | A | B | C | D | E |
| 28 | A | B | C | D | E |
| 29 | A | B | C | D | E |
| 30 | A | B | C | D | E |
| 31 | A | B | C | D | E |
| 32 | A | B | C | D | E |
| 33 | A | B | C | D | E |
| 34 | A | B | C | D | E |
| 35 | A | B | C | D | E |
| 36 | A | B | C | D | E |
| 37 | A | B | C | D | E |
| 38 | A | B | C | D | E |
| 39 | A | B | C | D | E |
| 40 | A | B | C | D | E |

|           |  |
|-----------|--|
| MARKS     |  |
| SECTION A |  |
| SECTION B |  |
| SECTION C |  |
| TOTAL     |  |

PAPER NUMBER

PERCENTAGE





$$\vec{R} = 6.52 \times 10^2 \text{ kmh}^{-1} \quad (1.81 \times 10^2 \text{ ms}^{-1}) \quad \text{W } 67.2^\circ \text{ S} \quad (1)$$

$$\Rightarrow \theta = 37.18^\circ \quad (1)$$

$$\frac{6.519 \times 10^2}{4.00 \times 10^2} = \frac{\sin 100^\circ}{\sin \theta}$$

$$= 1.811 \times 10^2 \text{ ms}^{-1}$$

$$= 6.519 \times 10^2 \text{ kmh}^{-1} \quad (1)$$

$$\vec{R} = \sqrt{(4.50 \times 10^2)^2 + (4.00 \times 10^2)^2 - 2(4.50 \times 10^2)(4.00 \times 10^2) \cos 100^\circ}$$

$$= 4.50 \times 10^2 \text{ kmh}^{-1} \text{ W } 30.0^\circ \text{ S} + 4.00 \times 10^2 \text{ S } 20.0^\circ \text{ E} \quad (1)$$

$$= 4.50 \times 10^2 \text{ kmh}^{-1} \text{ W } 30.0^\circ \text{ S} - 4.00 \times 10^2 \text{ N } 20.0^\circ \text{ W}$$

$$\Delta V = V - U$$

(c) Since friction is being ignored and the bullet is returning to the height it is fired, it will have the same magnitude of velocity.

$$V = 3.90 \times 10^2 \text{ ms}^{-1} \text{ downwards} \quad (1)$$

$$= 79.6 \text{ s} \quad (2)$$

$$\text{total time} = 2 \times 39.80$$

$$= 39.80 \text{ s}$$

$$\Rightarrow t = \frac{3.90 \times 10^2}{9.80}$$

$$\Rightarrow 0 = -3.90 \times 10^2 + (9.80)t$$

$$v = u + at$$

(b) Consider the time for half the flight.

$$\text{height} = 7.76 \times 10^3 \text{ m} \quad (2)$$

$$= -7.760 \times 10^3 \text{ m}$$

$$\Rightarrow s = \frac{19.6}{-1.521 \times 10^5}$$

$$\Rightarrow 0 = (-3.90 \times 10^2)^2 + 2(9.80)s$$

$$v^2 = u^2 + 2as$$

$$(\text{down} + v)$$

4. (a) Consider half of the flight.

$$v = 0 \text{ ms}^{-1}$$

$$u = -3.90 \times 10^2 \text{ ms}^{-1}$$

$$a = 9.80 \text{ ms}^{-2}$$

$$t = ?$$

$$s = ?$$

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$$1. \quad u = \frac{s}{t} = \frac{7.82 \times 10^{-2}}{0.0800} = 0.978 \text{ ms}^{-1} \quad (1)$$

$$v = \frac{s}{t} = \frac{1.75 \times 10^{-2}}{0.0800} = 0.219 \text{ ms}^{-1} \quad (1)$$

$$a = \frac{v-u}{t} = \frac{0.219 - 0.978}{0.180} = -4.22 \text{ ms}^{-2} \quad (1)$$

$\therefore \text{acceleration} = 4.22 \text{ ms}^{-2} \text{ backwards.} \quad (1) \quad (4)$

2. (a) AB: constant acceleration      CD: constant deceleration  
EF: constant velocity      FG: constant deceleration      (2)

(b)  $a = \frac{v-u}{t} = \frac{0 - 20.0}{2.00} = -10.0 \text{ ms}^{-2}$

$\therefore \text{greatest acceleration} = 10.0 \text{ ms}^{-2} \text{ backwards.} \quad (2)$

(c)  $s = \text{area under graph}$

$$= \frac{1}{2}(4.00)(10.0) + (3.00)(10.0) + \frac{1}{2}(1.00)(5.00) + (3.00)(5.00) + \frac{1}{2}(2.00)(15.0) + (6.00)(20.0) + \frac{1}{2}(2.00)(20.0)$$

$$= 2.225 \times 10^2 \text{ m.}$$

$\therefore \text{distance} = 2.22 \times 10^2 \text{ m} \quad (3)$

3. WHITFORDS      Forwards +ve.

$$v = 30.56 \text{ ms}^{-1} \quad (1) \quad a = \frac{v-u}{t} = \frac{30.56 - 0}{14.0} = 2.183 \text{ ms}^{-2}$$

$$u = 0 \text{ ms}^{-1}$$

$$a = ?$$

$$t = 14.0 \text{ s} \quad \therefore \underline{a = 2.18 \text{ ms}^{-2} \text{ forwards}} \quad (2)$$

$$s = ?$$

WARWICK

Forwards +ve

$$v = 0 \text{ ms}^{-1} \quad v^2 = u^2 + 2as$$

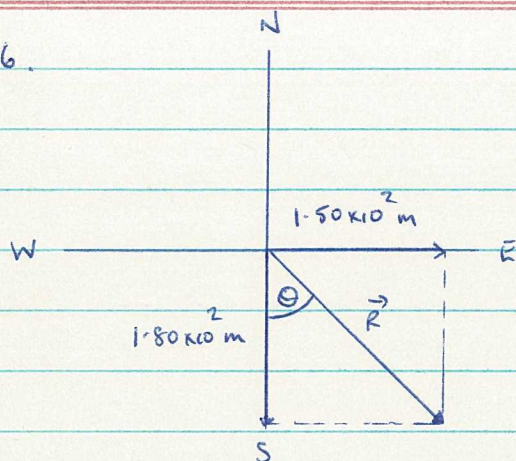
$$u = 30.56 \text{ ms}^{-1} \quad \Rightarrow 0 = (30.56)^2 + 2a(1.80 \times 10^2)$$

$$a = ? \quad \Rightarrow a = -2.594 \text{ ms}^{-2}$$

$$t = ?$$

$$s = 1.80 \times 10^2 \text{ m} \quad \therefore \underline{a = 2.59 \text{ ms}^{-2} \text{ backwards.}} \quad (2)$$

6.



$$(a) \quad \vec{R} = \sqrt{(1.50 \times 10^2)^2 + (1.80 \times 10^2)^2}$$

$$= 2.343 \times 10^2 \text{ m}$$

$$\tan \theta = \frac{1.50 \times 10^2}{1.80 \times 10^2}$$

$$\Rightarrow \theta = 39.81^\circ$$

$$\therefore \text{displacement} = 2.34 \times 10^2 \text{ m } S 39.8^\circ E \quad (3)$$

$$(b) \quad \text{speed} = \frac{\text{distance}}{\text{time}}$$

$$= \frac{3.30 \times 10^2}{54.0}$$

$$= 6.111 \text{ ms}^{-1}$$

$$\therefore \text{average speed} = 6.11 \text{ ms}^{-1} \quad (1)$$

$$(c) \quad v = \frac{s}{t}$$

$$= \frac{2.343 \times 10^2}{54.0}$$

$$= 4.339 \text{ ms}^{-1} \quad (2)$$

$$\therefore \text{average velocity} = 4.34 \text{ ms}^{-1} S 39.8^\circ E$$

TOTAL: 31